

Explainable and Efficient AI for Time Series Forecasting of Wind and Solar Energy in the TenneT Region, Germany

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Module: EEEM073 - AI and Sustainability

1. Introduction

Renewable energy sources such as wind and solar are the key players to move towards clean energy sources in the worldwide. But their generation capacity relies heavily on the prevailing weather, which means there will be fluctuations that affect load balancing within the power network. Accurate predictions are critical in helping avoid keeping fossil fuel-powered backup systems operating when they are not needed.

This study focuses on the TenneT DE control zone, covering 40% of German territory including wind-dominant Schleswig-Holstein and solar-dominant Bavaria. Wind onshore generation grew from 37 TWh in 2021 to 50 TWh in 2023, and solar PV from 17 TWh to 23 TWh, making accurate forecasting increasingly critical.

This study performs 1-hour-ahead forecasting, consistent with the standard operational horizon used by transmission system operators and published research reporting R^2 above 0.93 for this task (Karaman, 2023; Rajaperumal and Columbus, 2025).

ARIMA and other statistical techniques operate under the assumption of linearity and stationarity and do not take into account simultaneous effects from several exogenous weather parameters. The wind turbine output exhibits non-linearity, while the performance of solar energy depends on many interlinked factors, necessitating the use of AI-based methods capable of learning patterns from raw data. Three increasingly sophisticated algorithms are tested: Linear Regression as a base model, MLP Neural Networks for non-linear data, and LSTM networks.

In addition to sustainability in terms of AI, the study applies compression algorithms to decrease model size and prediction time, along with SHAP analysis, which helps identify the features that make predictions.

Contributions to SDG 7 (Affordable & Clean Energy) and SDG 13 (Climate Action), where our work can enhance accuracy in renewables' forecasting, as well as contribute to increased renewable usage and develop light-weight AI, whose carbon footprint is lower, consistent with the goals set by the German government of achieving 80% renewables electricity by 2030 and carbon neutrality by 2045.

1.1 Research Objectives

The objectives of this paper include: (1) developing an entire AI model for forecasting wind and solar production within the DE region under the TenneT; (2) comparing Linear Regression, MLP, and LSTM through their performance metrics MAE, RMSE, and R^2 ; (3) conducting the SHAP analysis; and (4) compression of all models.

1.2 Why AI is Needed

AI is required for three reasons: the non-linear physics of wind and solar systems, multi-scale temporal dependencies (hourly, daily, weekly), and the multivariate complexity of combining weather data from two locations. This level of multivariate, non-linear and temporal complexity makes traditional statistical approaches insufficient, thereby justifying the use of AI-based methods.

2. Dataset Description

2.1 Data Sources

Totally three open source datasets were used for the period from January 2021 to December 2024 included data for four years at an hourly frequency. The electricity production dataset was obtained from the SMARD website (Bundesnetzagentur, 2024) and contained information on the hourly electricity production measured in MWh for the region TenneT DE. There were Three target variables chosen are wind onshore, wind offshore, and solar PV.

The data on weather conditions were collected through the Open-Meteo ERA5 database (Zippenfenig, 2023) for two cities: Munich at 48.14°N latitude and 11.58°E longitude, which represents the south solar conditions, and Kiel at 54.32°N latitude and 10.12°E longitude, which represents the north wind conditions. Two cities are required as the TenneT DE region is too much extensive, it is covering more than 600 km from north to south.

Among the weather parameters used are 2m temperature, 10m and 100m wind speed, cloud cover, and shortwave radiation. This is due to the fact that wind speed is directly related to wind power generation, while shortwave radiation and cloud cover affect the generation of solar PV.

Table 1: Dataset Summary

Dataset	Source	Location	Period	Rows	Key Variables
Generation Data	SMARD (Bundesnetzagentur, 2024)	TenneT DE Zone	2021-2024	35,064	Wind Onshore, Wind Offshore, Solar PV (MWh)
Munich Weather	Open-Meteo ERA5 (Zippenfenig, 2023)	Munich, Bavaria	2021-2024	35,064	Temperature, Wind Speed 10m/100m, Solar Radiation, Cloud Cover
Kiel Weather	Open-Meteo ERA5 (Zippenfenig, 2023)	Kiel, Schleswig-Holstein	2021-2024	35,064	Temperature, Wind Speed 10m/100m, Solar Radiation, Cloud Cover

Reproducibility was ensured through manual extraction of generation data by specifying the TenneT DE control zone, one-hour temporal resolution and timeframe between January 2021 and December 2024 on the SMARD electricity market data portal. The chosen generation variables consisted of on-shore wind, off-shore wind, and solar PV and were saved under as generation_data.csv. Weather data were from Munich and Kiel and was acquired from the Open-Meteo Historical Weather API, which uses ERA5 variables including air temperature, wind speed at 10m and 100m, cloud cover, and shortwave radiation. The weather data files were called "munich_weather.csv.csv" and "kiel_weather.csv.csv". The three data files were loaded using Python from the "Data/" folder, matched by timestamps, corrected for any daylight savings issues, and combined to form one hourly file. The source is open access, and verification is possible by entering the date range, hourly frequency, variables, and locations shown in Table 1.

Although all-weather variables were collected for both locations, Kiel wind-speed features are most relevant for wind forecasting, while Munich shortwave-radiation and cloud-cover features are most relevant for solar forecasting.

2.2 Data Dictionary

Total record count = 34,896 with 34 independent variables and 3 dependent variables in the generated data. Missing 168 cases represent the starting week when lagged-168 variables cannot be calculated. Independent variables are listed in the table below.

Table 2: Data Dictionary - Key Features and Targets

Variable	Type	Unit	Description
wind_onshore_mwh	Target	MWh	Wind onshore generation in TenneT DE zone
wind_offshore_mwh	Target	MWh	Wind offshore generation in TenneT DE zone
solar_mwh	Target	MWh	Solar PV generation in TenneT DE zone
kiel_wind_speed_100m_km_h	Feature	km/h	Wind speed at 100m height in Kiel (northern zone)
munich_shortwave_radiation_w_m2	Feature	W/m2	Shortwave solar radiation in Munich (southern zone)
wind_onshore_mwh_lag1	Feature	MWh	Wind onshore generation 1 hour ago
wind_onshore_mwh_lag24	Feature	MWh	Wind onshore generation 24 hours ago (same hour yesterday)
wind_onshore_mwh_lag168	Feature	MWh	Onshore Wind generation 168 hours ago (same hour last week)
wind_onshore_mwh_rolling24	Feature	MWh	Wind onshore Average generation over the past 24 hours
hour_sin / hour_cos	Feature	Unitless	Cyclic encoding of hour of day
month_sin / month_cos	Feature	Unitless	Cyclic encoding of month of year

2.3 Data Split

To prevent leakage data was split chronologically three years for training, H1 2024 for validation and H2 2024 for testing (Table 3).

Table 3: Train/Validation/Test Split

Set	Period	Rows	Percentage
Training	Jan 2021 - Dec 2023	26,112	74.8%
Validation	Jan 2024 - Jun 2024	4,368	12.5%
Test	Jul 2024 - Dec 2024	4,416	12.7%
Total	Jan 2021 - Dec 2024	34,896	100%

Two separate MinMaxScalers were fitted on training data only - one for the 34 input features and one for the 3 generation targets - preventing data leakage and enabling clean inverse-transformation of predictions to MWh.

To minimise data leakage, all lag and rolling features were created using only past observations. Rather than randomly splitting this dataset was then split chronologically with 2021–2023 used for training, January–June 2024 for validation, and July–December 2024 for final testing. Feature and target scalers were fitted only on the training set and then applied to validation and test sets. No future target values from the validation or test periods were used during model training. Therefore, the high R^2 values mainly reflect the strong temporal persistence of 1-hour-ahead renewable forecasting rather than information leakage.

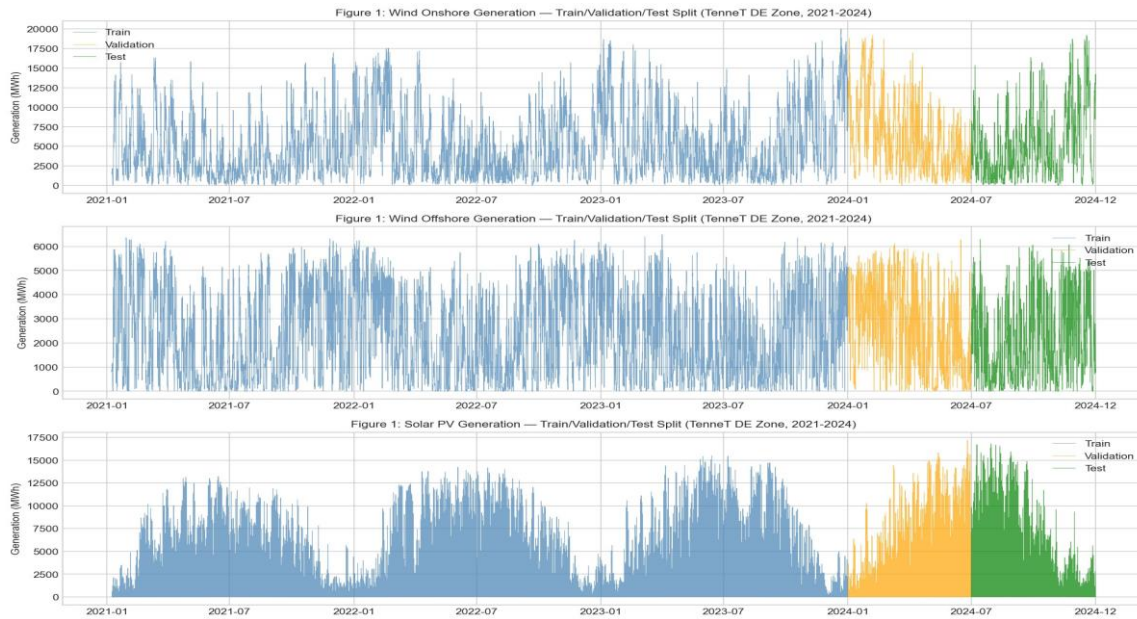


Figure 1: The chronological train/validation/test split was represented in three colors. The blue color displays the training period from 2021 to 2023, while the orange color displays the validation period from Jan to Jun 2024, whereas the green color displays the test period from Jul to Dec 2024. The clear color boundaries confirm no overlap between sets, which is essential for honest model evaluation.

3. Data Preprocessing

3.1 Data Cleaning

All three datasets loaded with zero missing values. Two issues were resolved: (1) comma thousands-separators in the generation data required correct parsing parameters; (2) eight Daylight Saving Time transitions created one-hour gaps in March and duplicate timestamps in October each year. Duplicates were removed and gaps filled by forward-filling the preceding hour's value, preserving lag feature integrity.

3.2 Feature Engineering

Three types of features were engineered, giving 34 input features in total.

Lag Features

Lag features at $t-1$, $t-24$ and $t-168$ were created for each target, capturing short-term momentum, the daily cycle and weekly seasonality. These specific lags were confirmed by ACF analysis in Section 5.

Rolling Mean Features

24-hour and 168-hour rolling averages were calculated for each target to provide the model with a smoothed trend signal, complementing the point-in-time lag features.

Cyclic Time Features

Time of the day, month and day of the week were encoded by means of sine and cosine (for example, $\text{hour_sin} = \sin(2\pi \times \text{hour} / 24)$), ensuring circular continuity by making hours 23 and 0 considered neighbors, rather than being opposites.

3.3 Descriptive Statistics

Table 4 presents a summary of the generation targets across the three generations. Onshore wind is the most volatile generation source (standard deviation 4,198 MWh), whereas solar generation has a small median of 67 MWh due to 3,693 nighttime zero-production periods, although its peak is 17,200 MWh during summer.

Table 4: Descriptive Statistics - Generation Targets (2021-2024)

Statistic	Wind Onshore (MWh)	Wind Offshore (MWh)	Solar PV (MWh)
Mean	5,139	2,322	2,349
Std Dev	4,198	1,717	3,644
Minimum	4.75	0.00	0.00
Median	3,892	2,116	67
Maximum	20,024	6,490	17,200
Zero-generation hours	0	0	3,693

4. Exploratory Data Analysis

4.1 Annual Generation Trends

Figure 2 indicates, annual generation totals. Wind onshore dominates, growing from 37 TWh (2021) to 50 TWh (2023). Solar PV grows steadily from 17 TWh to 23 TWh. While this upward renewable trend increases the consequences of forecast errors, directly motivating this study.

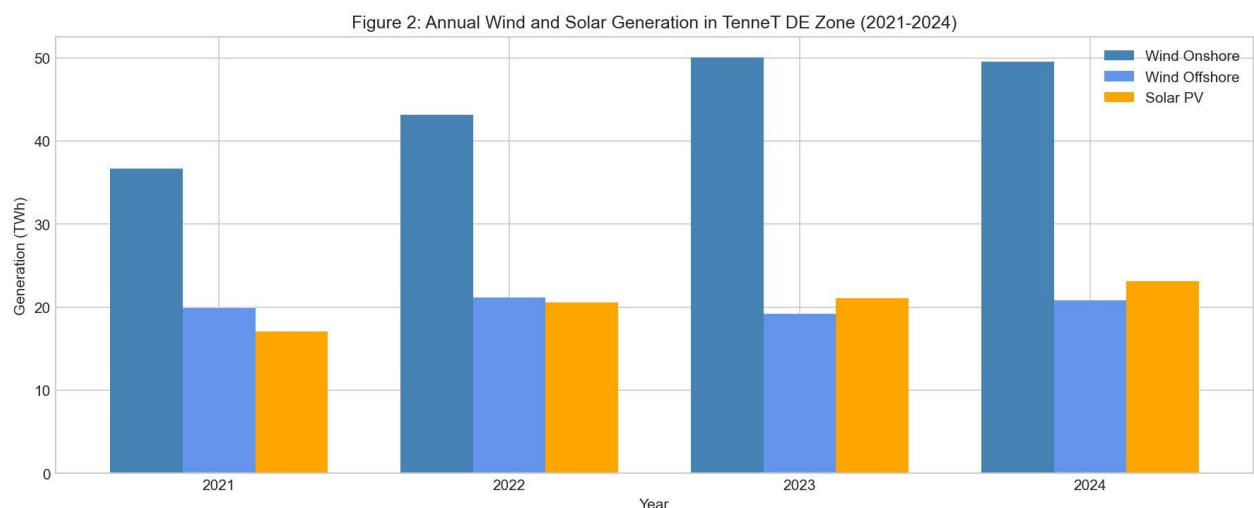


Figure 2: Annual generation totals by energy type. Wind onshore dominates and grows steadily over the period, while solar PV increases year on year. This confirms the growing importance of accurate renewable forecasting as fossil fuel backup capacity is reduced.

4.2 Seasonal and Hourly Patterns

As shown by Figure 3, wind energy and solar energy production exhibit inverse seasonal patterns, as the highest production of wind occurs between December and February because of storm formation in the Atlantic Ocean, while the highest production of solar occurs in June and July.

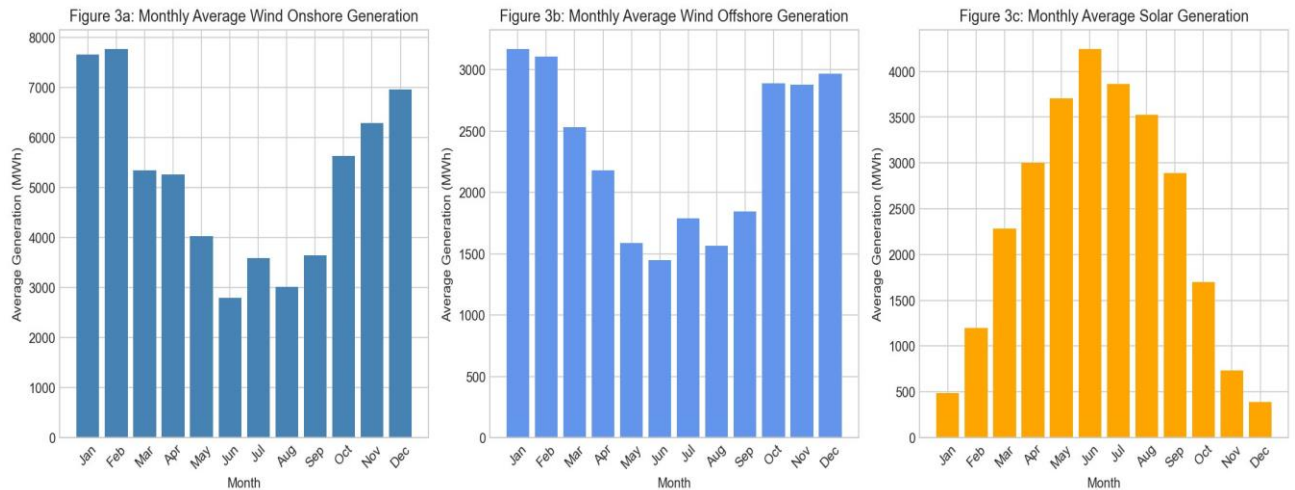


Figure 3: Monthly average generation showing the opposing seasonal patterns. Wind peaks in winter and solar peaks in summer. This confirms why month-of-year encoding is important for all three forecasting targets, and why the training data covering multiple full years is needed to capture these seasonal effects.

Figure 4 shows the hourly profile. Solar follows a clear bell curve peaking at noon. Wind shows a subtler daily pattern with a mid-morning dip. Both confirm that hour of day is an important predictor requiring cyclic encoding.

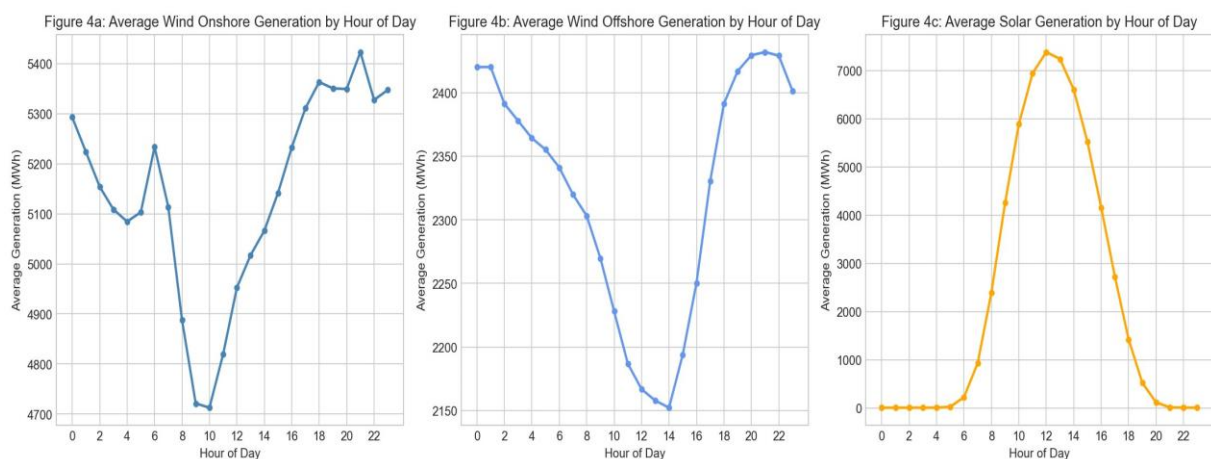


Figure 4: Average generation by hour of day. Solar which shows a clear bell-shaped daily cycle with a sharp peak around noon. Wind shows a smaller but consistent daily variation. Both patterns confirm that hour of day is a key predictor, justifying the inclusion of `hour_sin` and `hour_cos` as features.

4.3 Distribution Analysis

Figure 5 shows boxplots of the three targets. Wind onshore has the widest spread with high-value outliers from storm events; solar is extremely right-skewed due to night-time zeros. These distributions confirm MinMaxScaling over z-score standardisation.

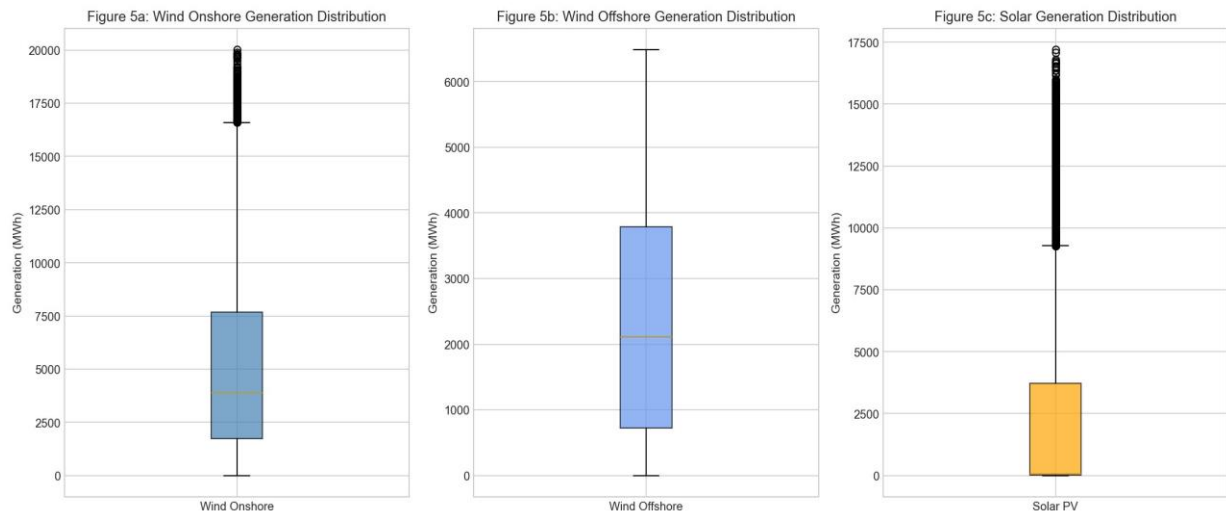


Figure 5: Boxplot distributions of all three targets. Wind onshore shows wide spread and high-value outliers from storm events. Solar shows extreme right-skew due to night-time zeros. Both distributions confirm that MinMaxScaling is more appropriate than standardisation for this dataset.

4.4 Correlation Analysis

Figure 6 displays the correlation matrix. The wind speed in Kiel at 100m is highly correlated with the wind generation ($r = 0.83$), while short wave radiation at Munich is highly correlated with solar ($r = 0.91$), justifying the selection of sites.

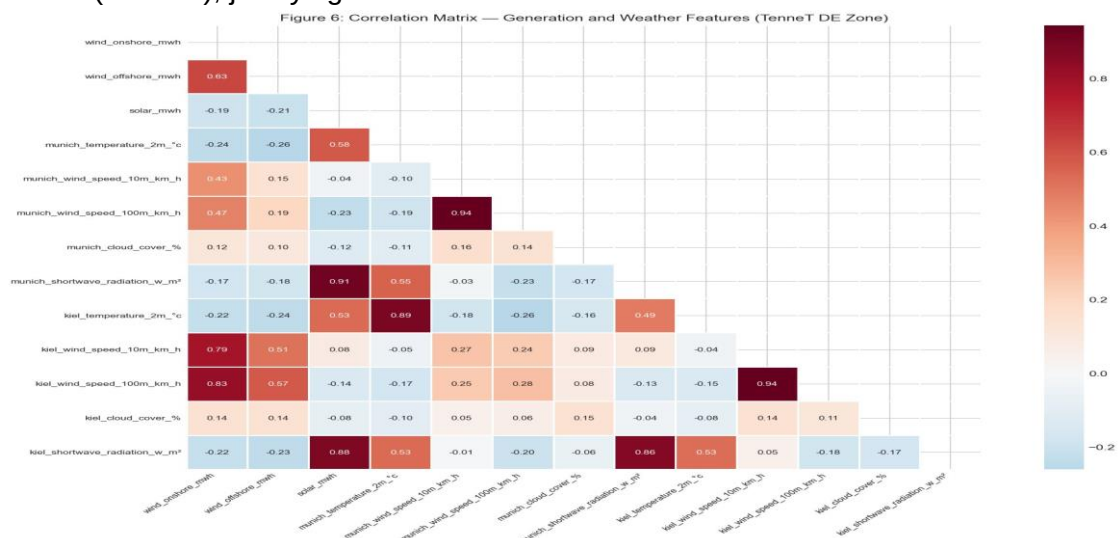


Figure 6 Pearson correlation matrix. The important results are: wind speed of Kiel at 100m is highly correlated with wind energy generation ($r=0.83$), confirming the right location choice; solar radiation of Munich is highly correlated with solar energy generation ($r=0.91$), confirming the right south-oriented location choice; wind and solar energy production have a negative correlation ($r=-0.19$).

4.5 Weather vs Generation Scatter Plots

Non-linear associations can be observed in Figure 7. However, the association for the generation by Wind is cubic because the power generated is constant after a certain speed of wind. The association of solar generation exhibits fan-shaped patterns due to the influence of clouds.

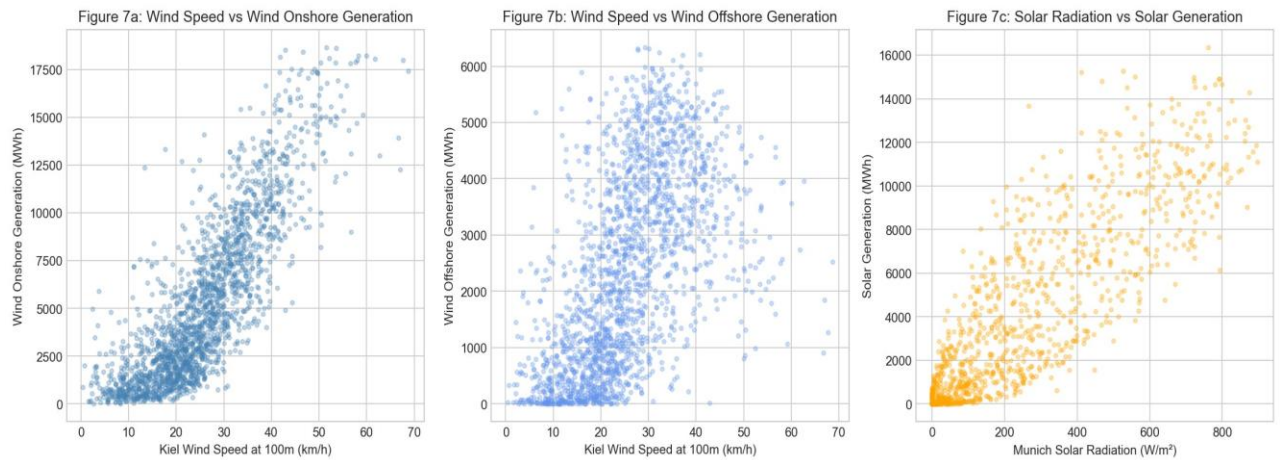


Figure 7: Scatter plot between different weather variables and generation targets. The non-linear association of wind speed to wind generation target (wind power curve) and non-linear relationship between scattered clouds and solar radiation further establish the need for non-linear models over a linear benchmark model.

4.6 Summer vs Winter Contrast

Figure. 8 shows the comparative graph between winter and summer seasons. The generation of wind power is roughly 2.5 times greater in winter than in summer, whereas the solar power achieves its maximum output at 11,000 MWh in summer as compared to 3,000 MWh in winter.

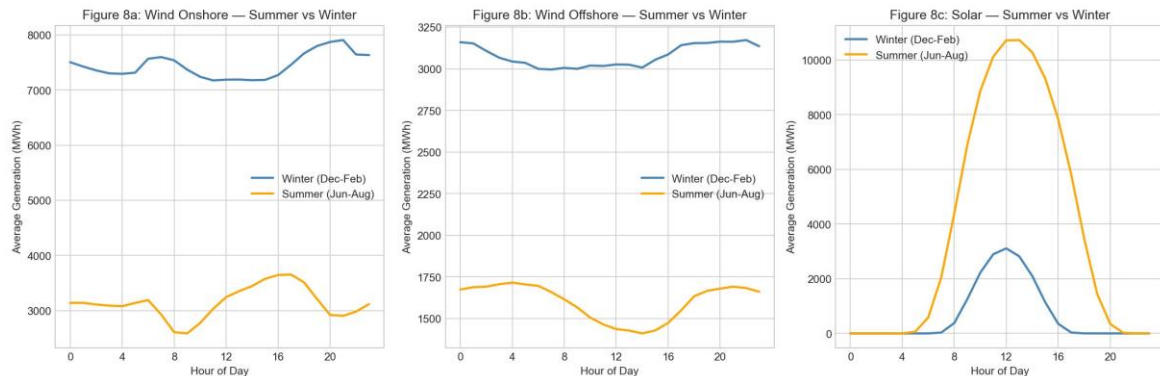


Figure 8: Summer versus winter generation profiles. Wind is approximately 2.5 times higher in winter while solar is approximately 3.7 times higher in summer. This confirms why both month encoding and long-horizon rolling features are essential - the model must adapt its predictions based on the current season.

5. Time Series Analysis

5.1 Autocorrelation Analysis (ACF)

In Figure 9, the ACF is presented for the first 200 lags. The presence of slow decay of autocorrelation in wind data signifies persistence over many days. Solar data on the other hand displays spikes in its autocorrelation at each day, corresponding to the daily cycle.

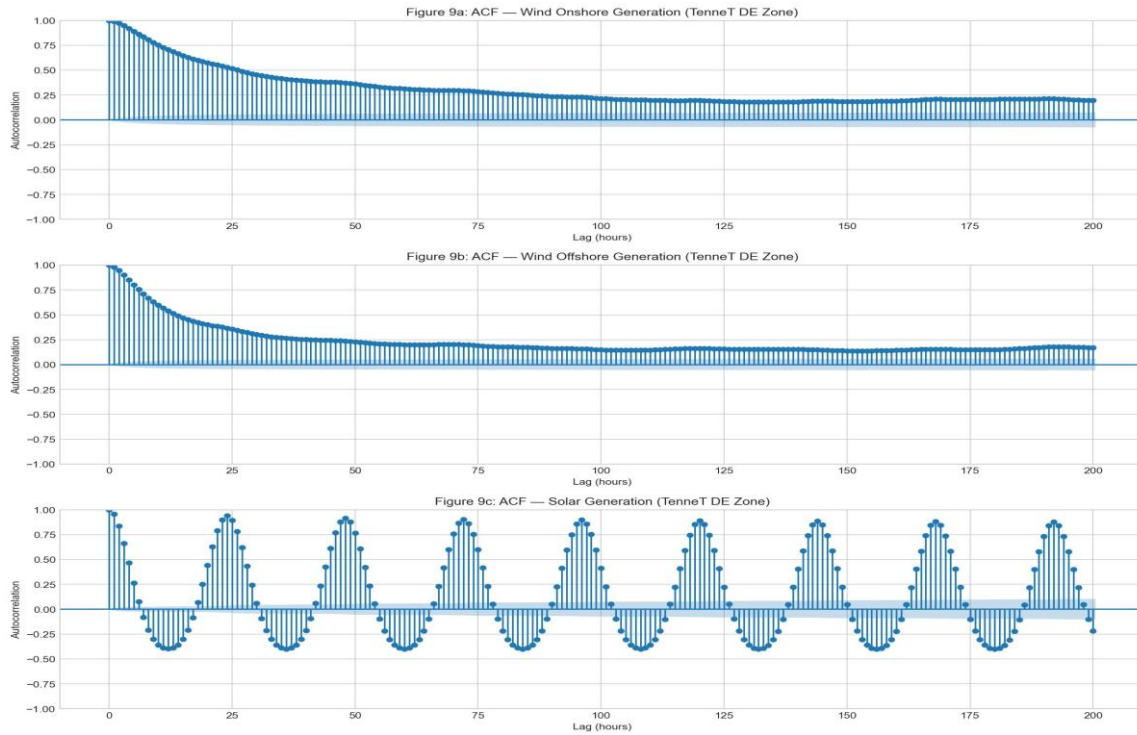


Figure 9: ACF plots up to 200 lags. Wind shows slowly decaying autocorrelation confirming multi-day temporal memory. Solar shows clear peaks every 24 hours confirming the daily cycle. The significant values at lags 1, 24 and 168 directly justify the lag features selected in the preprocessing stage, confirming these were the right choices.

5.2 Partial Autocorrelation Analysis (PACF)

Figure 10 shows the PACF. Wind shows a dominant lag-1 spike followed by a sharp cutoff, confirming that the previous hour is the single most important direct predictor. This sharp cutoff validates using only three targeted lags (t-1, t-24, t-168), keeping the model lightweight.

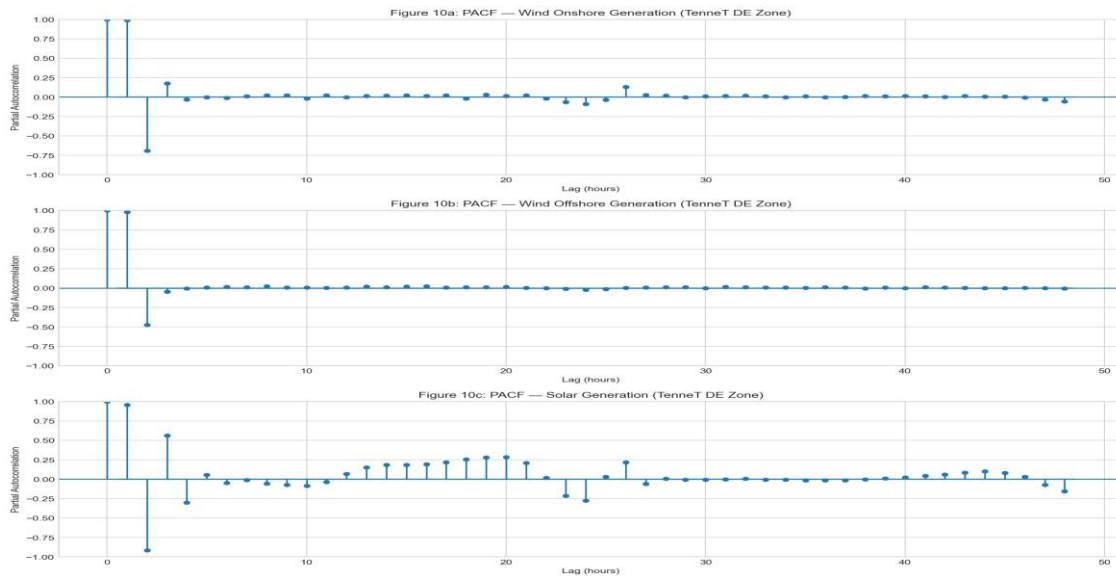


Figure 10: Plots for the partial autocorrelation function (PACF) illustrating how lag-1 is dominant for wind, with a sharp cut-off thereafter, whereas solar energy exhibits further structure at lags 18 to 25. This sharp cut-off demonstrates that using three well-chosen lags is adequate, without requiring hundreds of lagged features.

5.3 STL Seasonal Decomposition

Figures 11 and 12 show Decomposition of STL (Jan-Jun 2021). Regarding wind power (see Figure 11), the weekly seasonal cycle is around $\pm 6,000$ MWh, confirming that the t-168-time lag is valid. The very high value of the residual (over 13,000 MWh

For solar (Figure 12), the trend rises from near zero in January to 3,500 MWh by June, and the residual is much smaller relative to the seasonal component, confirming solar is more predictable than wind - consistent with the higher R^2 scores achieved for solar in Section 7.

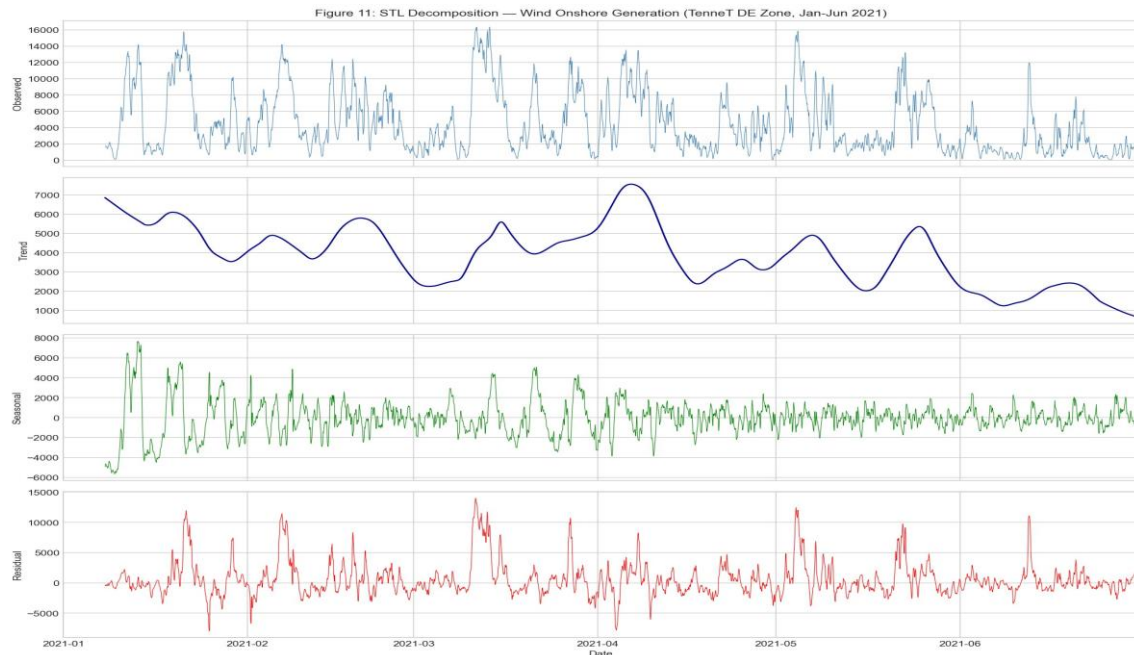


Figure 11: STL decomposition of wind onshore generation. The trend shows a gradual seasonal decline, the seasonal component confirms a weekly pattern of approximately $\pm 6,000$ MWh validating the t-168 lag feature, and the large residual confirms that significant wind variability is inherently unpredictable - setting a practical accuracy ceiling for all forecasting models.

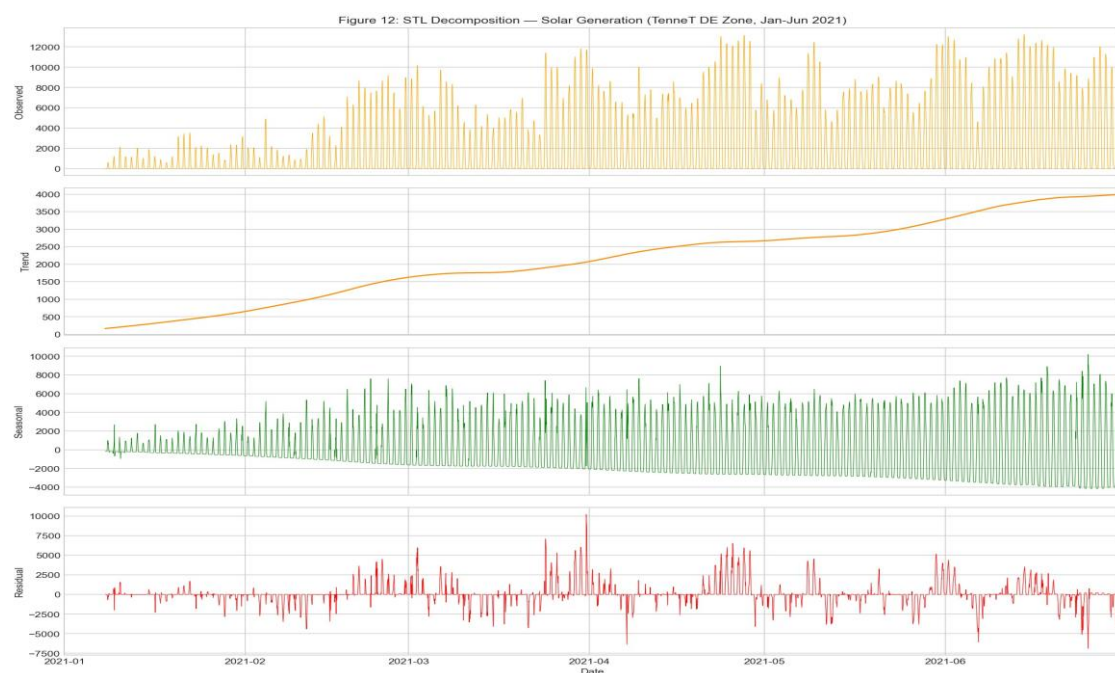


Figure 12: STL decomposition of solar generation. The clear upward trend reflects lengthening days, the regular seasonal component reflects the daily cycle, and the small residual confirms that solar is more predictable than wind. This directly predicts that all models should achieve higher R2 for solar - a prediction confirmed by the results in Table 8.

5.4 Stationarity Testing

Table 5 whereas shows the Augmented Dickey-Fuller stationarity test results. All three series return p-values near zero, indicating that the main statistical structure is sufficiently stable for chronological model evaluation. This justifies using 2021–2023 data for training and 2024 data for testing/validation, despite the possibility of operational variations, climatic anomalies, and capacity variations leading to non-stationarity. Thus, stationarity must not be seen as an affirmation but rather as proof of generalisation.

Table 5: Augmented Dickey-Fuller Stationarity Test Results

Series	ADF Statistic	p-value	Result
Wind Onshore	-14.3283	0.000000	Stationary (confirmed)
Wind Offshore	-17.2568	0.000000	Stationary (confirmed)
Solar PV	-6.9529	0.000000	Stationary (confirmed)

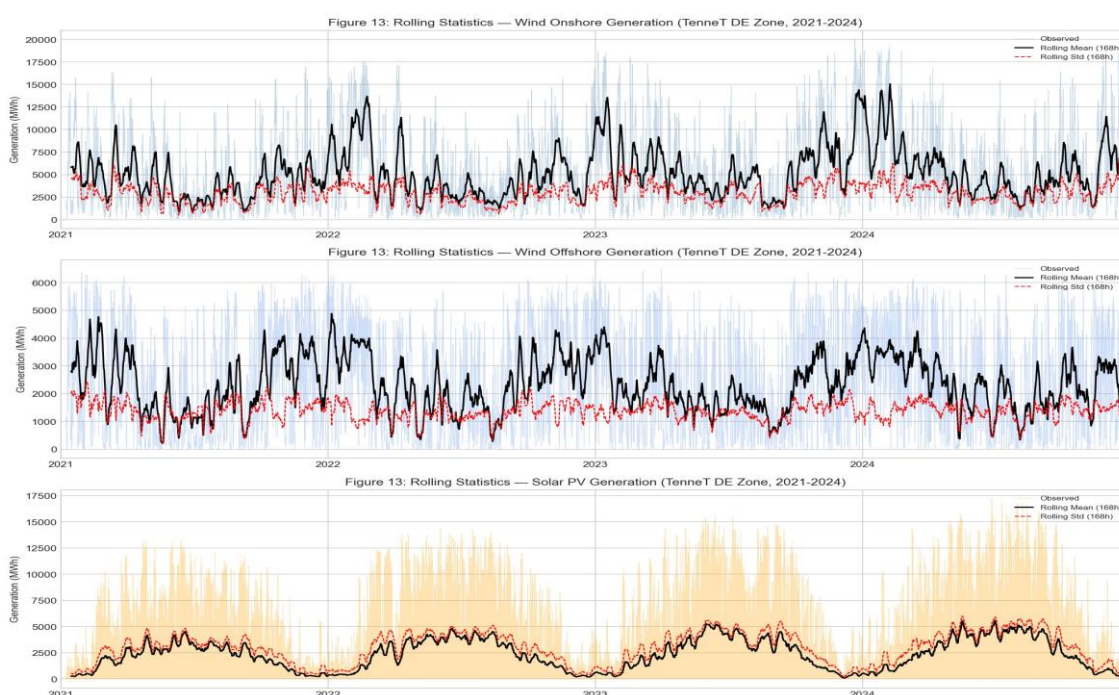


Figure 13: Rolling mean for 168 hours (black) and rolling standard deviation (red dotted). Neither of these two lines demonstrates any sort of tendency towards growth or decline and looks constant. This provides additional confirmation for stationarity test results based on ADF tests. Nevertheless, this does not imply future generalization guarantees at all.

6. AI Modelling

6.1 Model Selection

A total of Three supervised models of increasing complexity were selected, every one of them representing a different trade-off between accuracy, interpretability and computational cost.

6.1.1 Linear Regression (Baseline)

Linear Regression fits a straight-line relationship between the 34 features and three targets. It is extremely fast (0.14s training, 1.64 KB), fully interpretable and serves as the baseline. Its limitation is the inability to capture non-linear relationships, though the engineered lag features compensate significantly.

6.1.2 Multi-Layer Perceptron (MLP)

The MLP treats each data point separately without any time-memory component. The network architecture is comprised of three hidden layers having ReLU activation function and Dropout (0.2). For training, Adam along with Early Stopping and ReduceLROnPlateau was employed and the process took 36 seconds to complete.

Table 6: MLP Architecture

Layer	Type	Units	Activation	Parameters
Input	Dense	128	ReLU	4,480
Dropout	Dropout (0.2)	--	--	0
Hidden 1	Dense	64	ReLU	8,256
Dropout	Dropout (0.2)	--	--	0
Hidden 2	Dense	32	ReLU	2,080
Output	Dense	3	Linear	99
Total	--	--	--	14,915

6.1.3 LSTM (Long Short-Term Memory)

Long Short-Term Memory utilizes internal memory throughout the time steps via gating. A 24-hour lookback window was used ($24 \times 34 = 816$ values per sample), motivated by the strong lag-24 autocorrelation in Figure 9. Two stacked LSTM layers (128 and 64 units) allow hierarchical temporal learning (Table 7). Training completed in 671 seconds.

Table 7: LSTM Architecture

Layer	Type	Units	Parameters
Input	LSTM (return sequences)	128	83,456
Dropout	Dropout (0.2)	--	0
Hidden	LSTM	64	49,408
Dropout	Dropout (0.2)	--	0
Dense	Dense (ReLU)	32	2,080
Output	Dense (Linear)	3	99
Total	--	--	135,043

6.2 Training Results

Figure 14 represents the training history for the MLP. It is evident that the training and validation losses decrease smoothly and converge, thereby proving that there is no case of overfitting. Training for the MLP was stopped at epoch 16 by EarlyStopping.

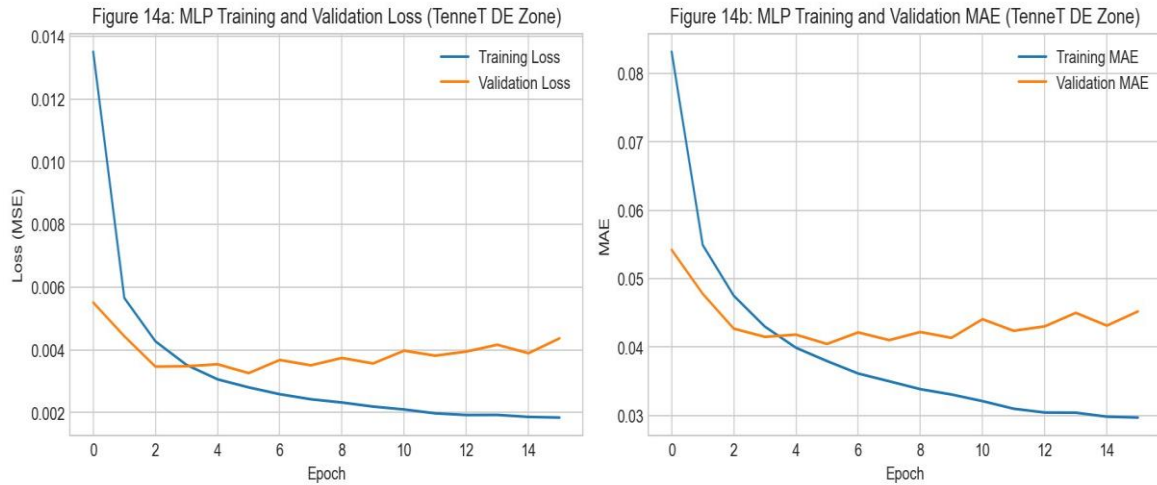


Figure 14: MLP training history over 16 epochs. Training and validation curves decrease together and converge closely, confirming no overfitting. The Dropout and EarlyStopping mechanisms worked as intended to produce a well-generalised model that will perform reliably on the unseen test set.

Figure 15 shows the LSTM training history, with similar smooth convergence over 22 epochs. The longer training time (671s vs 36s for MLP) is the primary sustainability cost of LSTM.

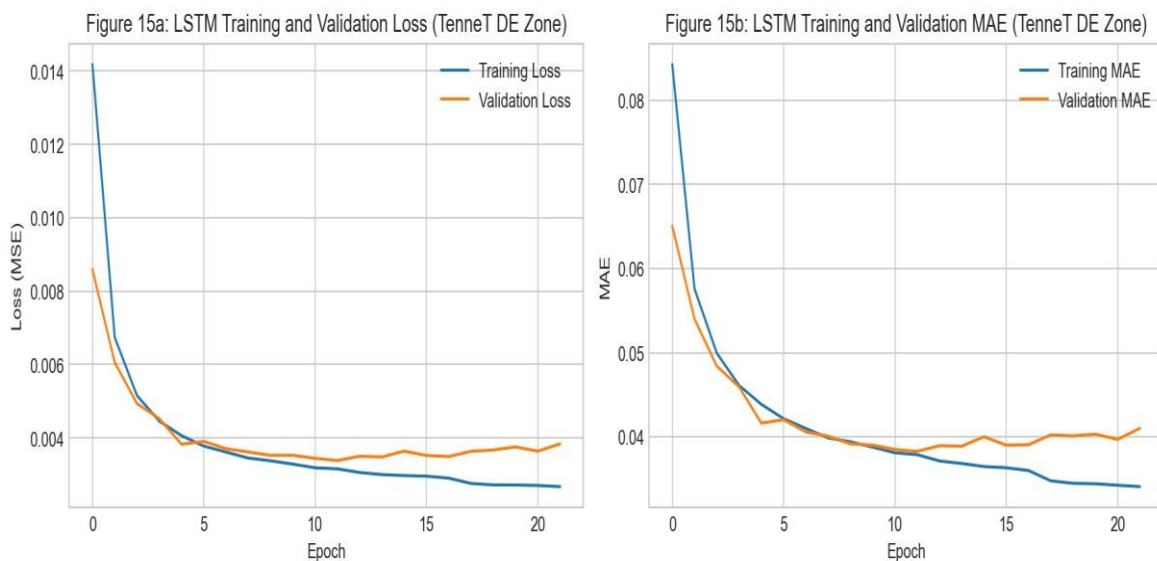


Figure 15: LSTM training history over 22 epochs showing consistent convergence of both training and validation curves. No overfitting is present, confirming that the larger LSTM model has been effectively regularised. The longer training time compared to MLP is the primary computational sustainability cost of using LSTM.

The graph in Figure 16 is the forecast vs actual output in the first week of July 2024. The Linear Regression model forecasts the trend of the wind pattern quite accurately while the LSTM model predicts solar quite accurately (as shown by $R^2 = 0.9821$).

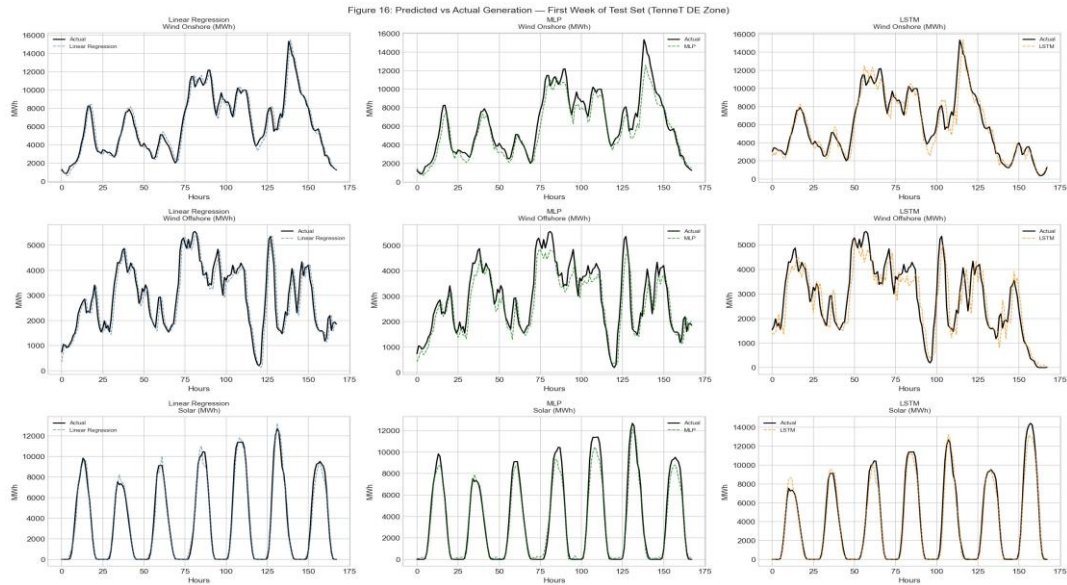


Figure 16: Prediction vs. Actual Generation for the Week of the First of July 2024. There is a very good fit between the Linear Regression Model (dashed blue line) and the generation trend overall with little to no deviation. There is a better accuracy of the prediction of the generation trend using the LSTM model (dashed orange line).

7. Model Evaluation and Results

7.1 Evaluation Metrics

MAE calculates the mean absolute value of prediction errors measured in MWh. RMSE gives greater weight to larger prediction errors, especially in cases where such errors can negatively impact grid stability. R^2 represents the amount of variability that is accounted for by the model, which means the higher the R^2 value, the better the prediction

7.2 Quantitative Results

Table 8 shows the complete results. It is important to point out that Linear Regression obtains the best R^2 scores both for wind onshore (0.9845) and for wind offshore (0.9477), proving that with good feature engineering, one can beat deep learning techniques.

Table 8: Model Evaluation Results - Test Set (Jul-Dec 2024)

Model	Target	MAE (MWh)	RMSE (MWh)	R^2
Linear Regression	Wind Onshore	378.67	534.50	0.9845
Linear Regression	Wind Offshore	252.32	367.41	0.9477
Linear Regression	Solar PV	352.52	585.72	0.9792
MLP	Wind Onshore	721.19	994.49	0.9462
MLP	Wind Offshore	295.91	401.84	0.9374
MLP	Solar PV	443.39	733.19	0.9674
LSTM	Wind Onshore	611.15	843.70	0.9614

Model	Target	MAE (MWh)	RMSE (MWh)	R2
LSTM	Wind Offshore	385.83	540.87	0.8871
LSTM	Solar PV	302.53	543.04	0.9821

Table 9: Model Size and Computational Efficiency

Model	Size (KB)	Training Time (s)	Inference Time (ms)
Linear Regression	1.64	0.14	1
MLP	208.54	36.45	718
LSTM	1,623.37	671.10	2,571

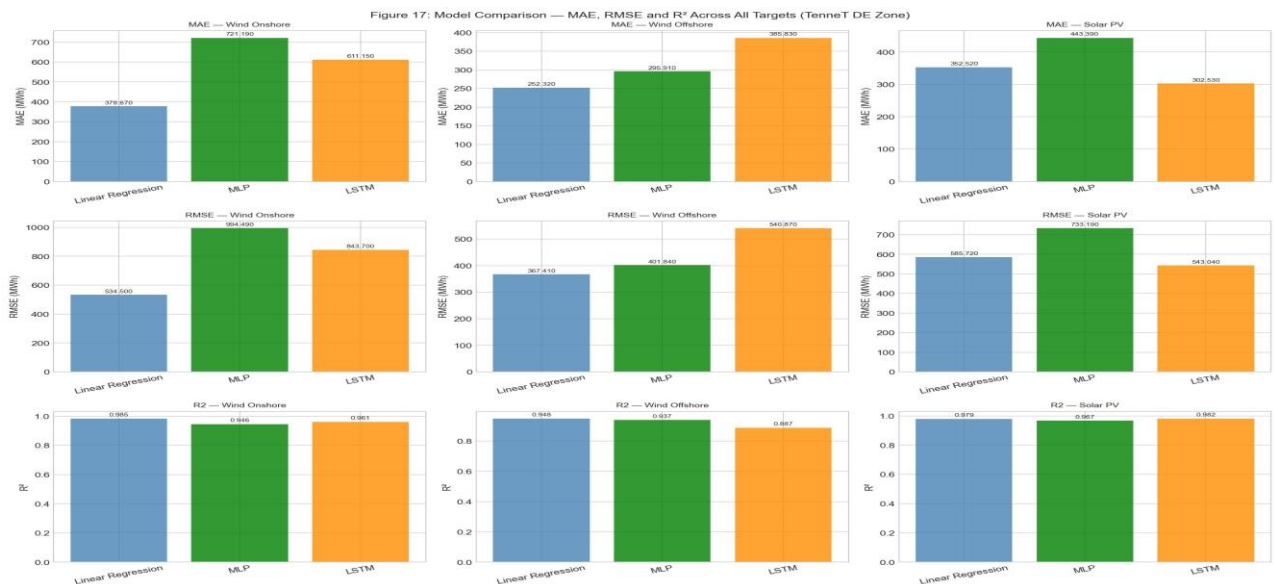


Figure 17: shows that Bar chart comparing all three models across all three targets. In the MAE and RMSE rows, shorter bars mean lower to minimal error (better performance). In the R² row, taller bars mean better fit. Linear Regression shows the shortest error bars for wind targets, confirming its strong wind performance. LSTM shows strong solar performance ($R^2 = 0.9821$), confirming its sequential memory advantage for solar prediction. Linear Regression achieves the highest R² for wind targets.

7.3 Discussion

These three models have an R² value greater than 0.88, which is similar to other published findings, as follows: R²=0.969 for LSTM wind forecast model as per Karaman (2023); R²=0.986 for hourly renewable energy forecast model as per Hossain et al. (2025); and R²=0.995 for wind energy forecast

The strong performance of Linear Regression is the most notable result. Once lag features linearise the temporal structure, neural network complexity provides limited additional benefit for wind forecasting under the current feature-engineered setting. Linear Regression is approximately 990 times smaller than LSTM and far faster, yet achieves higher accuracy for most targets — a compelling sustainable AI finding.

LSTM achieves the best solar performance ($R^2 = 0.9821$, $MAE = 302.53$ MWh). The 24-hour lookback window gives LSTM implicit access to the previous day's pattern, directly exploiting the lag-24 ACF peak.

Wind offshore shows the lowest R^2 (0.89–0.95) across all models, consistent with the large STL residual in Figure 11 — confirming inherent unpredictability beyond weather and time features.

At the 1-hour ahead horizon, temporal persistence (especially lag-1) dominates, making the problem approximately linear and limiting LSTM's advantage. The high R^2 values should therefore be interpreted in the context of the 1-hour-ahead forecasting horizon. Because lag features use only past observations, especially the previous hour and previous day, they are valid predictors rather than data leakage. The chronological train/validation/test split and fitting of scalers only on the training data further reduce leakage risk. In a real-world deployment using forecasted weather inputs, plant-level data, or longer prediction horizons, accuracy would likely be lower. For horizons that extend for periods when lagged variables lose significance (for instance, one-day forward), it is clear that deep learning will add more value. It is evident that the choice of model has to match the forecasting horizon.

Deep learning models become more valuable under conditions where temporal dependencies are complex, long-range, or difficult to explicitly engineer. This includes longer forecasting horizons, highly volatile weather conditions, or real-time forecasting scenarios using raw sequential inputs without manually engineered lag features.

One of the key learnings from the above experiment is that the performance of any predictive algorithm depends on the quality of feature engineering rather than model complexity. The use of lagged features in the dataset makes the entire problem linear and results in a simpler model like Linear Regression beating the deep learning model in performance while consuming substantially less computational power. The idea of sustainable AI, as discussed earlier, is that the best answer might not always be the one involving complex modeling.

Even though deep learning models do not surpass the performance of Linear Regression in this particular case of 1-hour prediction, they are crucial when it comes to more sophisticated situations where temporal relationships cannot be easily engineered.

This shows that choosing an appropriate model for sustainable AI depends on structure and efficiency rather than complexity, implying that simple models may be optimal if there is good feature engineering.

7.4 Residual Analysis

Figure 18 shows residual distributions for all models and targets. All distributions are centred near zero, confirming no systematic bias. Linear Regression produces the narrowest wind residuals and LSTM the narrowest solar residuals, consistent with Table 8. Wider wind tails reflect inherent unpredictability.

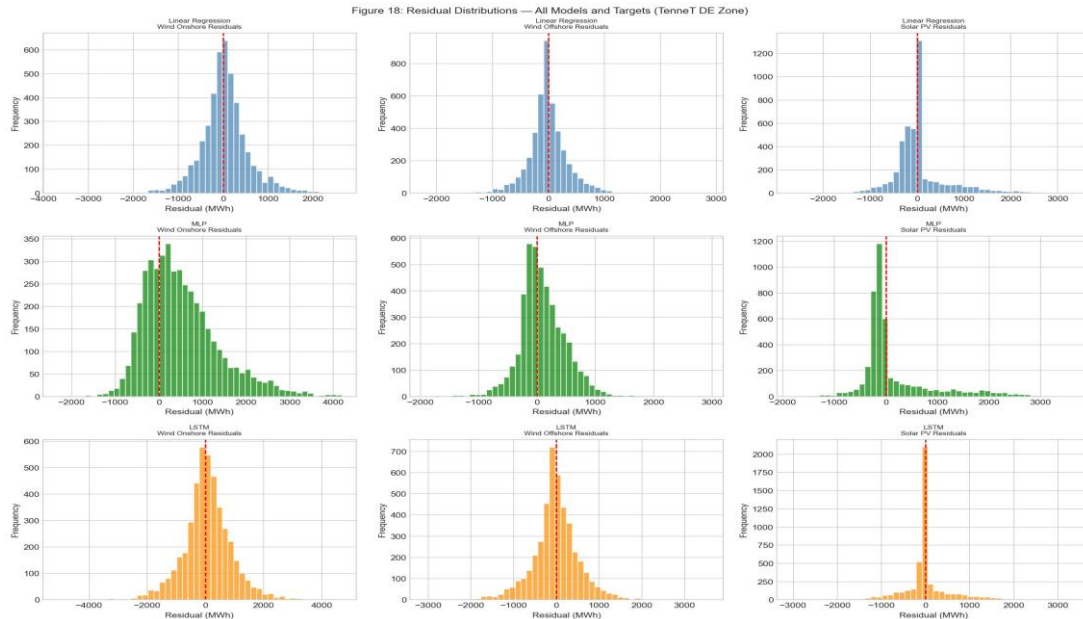


Figure 18: Residual histograms for all model-target combinations. All distributions are centred near zero confirming no systematic prediction bias. Linear Regression shows the tightest wind residuals and LSTM shows the tightest solar residuals, consistent with the R^2 values in Table 8. Wider wind residual tails across all models confirm the inherent unpredictability identified in the STL analysis.

8. Explainable AI - SHAP Analysis

SHAP (SHapley Additive exPlanations) was also applied to the three models to identify which variables impact the decision-making process. This step is crucial since grid managers should be able to know why such decisions are made.

8.1 Linear Regression SHAP

Figure 19 illustrates SHAP of Linear Regression. The lag-1 variables outweigh the other targets significantly, thus reiterating the results obtained from ACF. Wind speed in Kiel emerges as an important secondary factor in the case of wind and radiation in Munich for solar.

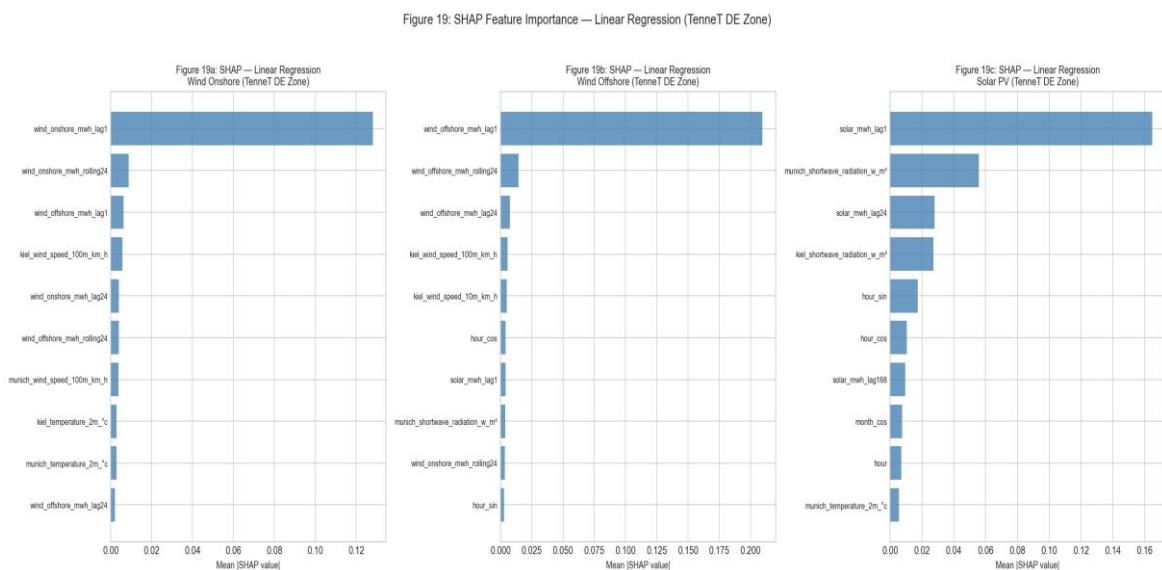


Figure 19: SHAP Feature Importance Chart for Linear Regression Model. Lag-1 features have completely dominated across all three target variables, suggesting that

the physics of the situation have been accurately captured by the model. Weather variables (Kiel wind speed for wind targets, Munich irradiation for solar targets) are shown as the second important variables, reinforcing the dual location weather variable strategy.

8.2 MLP SHAP

Figure 20 indicates SHAP for MLP. The lag-1 variables still remain more significant, proving that it is indeed a physical pattern. Moreover, MLP gives more significance to time variables as well as hour_sin and hour_cos because of its ability to recognize non-linear relationships unlike linear regression.

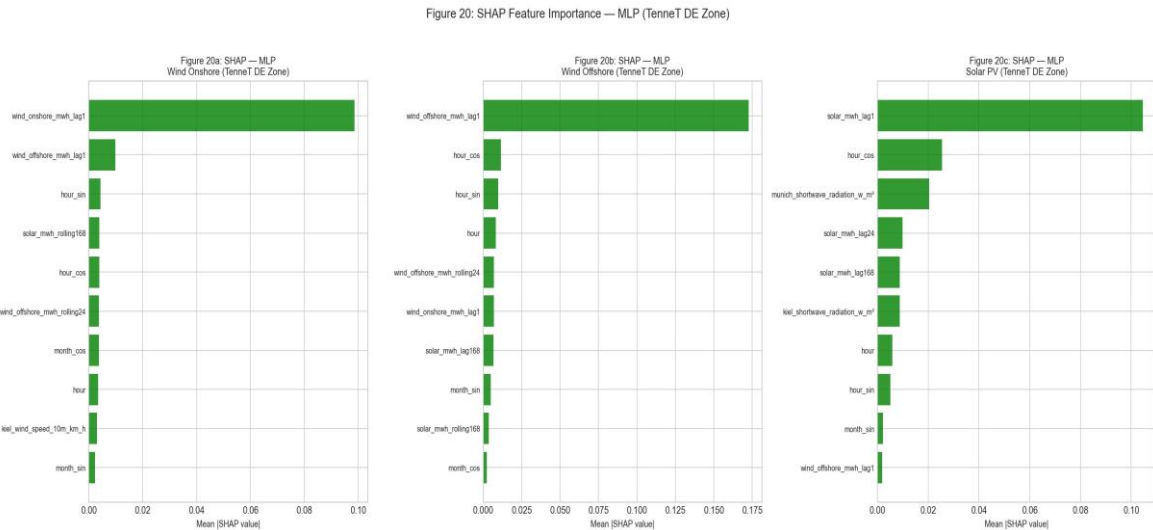


Figure 20: Feature importance based on SHAP values of MLP. The lag-1 variables again exhibit consistent domination in comparison with Linear Regression, thereby proving that both models have identified the same leading predictor. However, in addition to this, the MLP also places high importance on cyclic time-related variables (hour_cos and hour_sin).

8.3 LSTM SHAP

Figure 21 demonstrate SHAP values for LSTM (normalized relative importance). The features from various time lags over the 24-hour window are seen at the higher ranks, indicating that LSTM indeed makes use of its sequence memory and not only the last time lag.

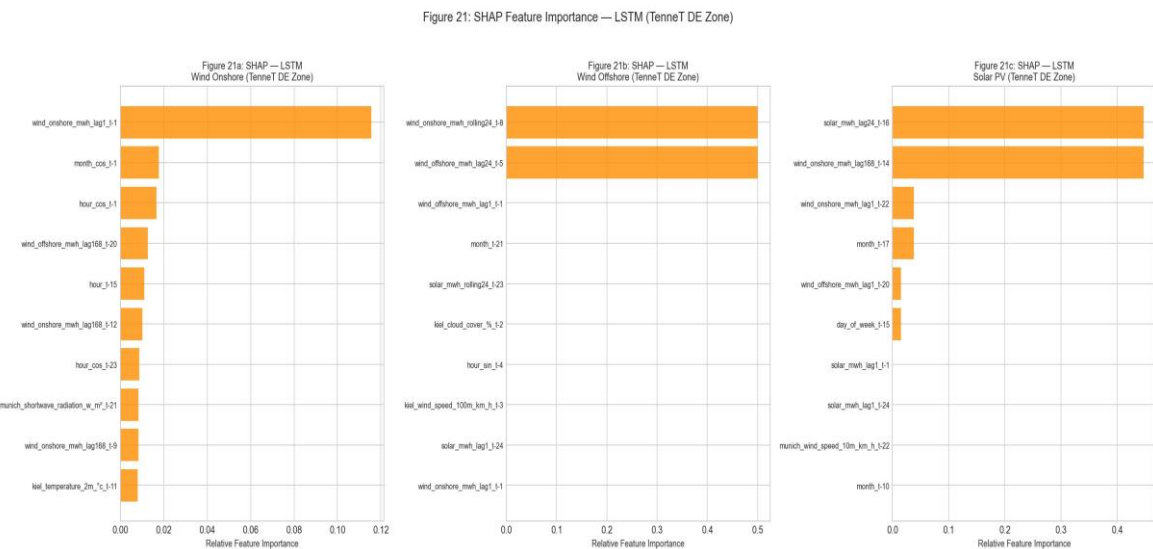


Figure 21: Importance of features in the LSTM model using SHAP explained through normalised relative importance values. Features belonging to different time steps in

the 24 hours look back period feature in the list of important features, indicating the fact that the LSTM model does indeed make use of its sequential memory capability.

8.4 LIME Local Explainability

SHAP gives the importance of each feature over the global level for all the predictions. LIME (Local Interpretable Model-Agnostic Explanations) fills the gap left by SHAP by giving the explanation for individual predictions on what features have pushed them higher or lower. Two such instances have been considered: the topmost solar generation hour from test data (2024-07-30 12:00, Actual: 16,796 MWh, Predicted: 15,933 MWh) and the topmost wind generation hour (2024-12-18 17:00, Actual: 19,195 MWh).

Figure 22 demonstrates the LIME explanations for Linear Regression. In Figure 22a, solar_mwh_lag1 is dominant with high weight (0.29), which proves that indeed the model has understood recent solar generation to be the key variable. The contribution of Munich shortwave radiation is positive, hence, proving the choice of location is correct. For wind generation instance shown in Figure 22b, the

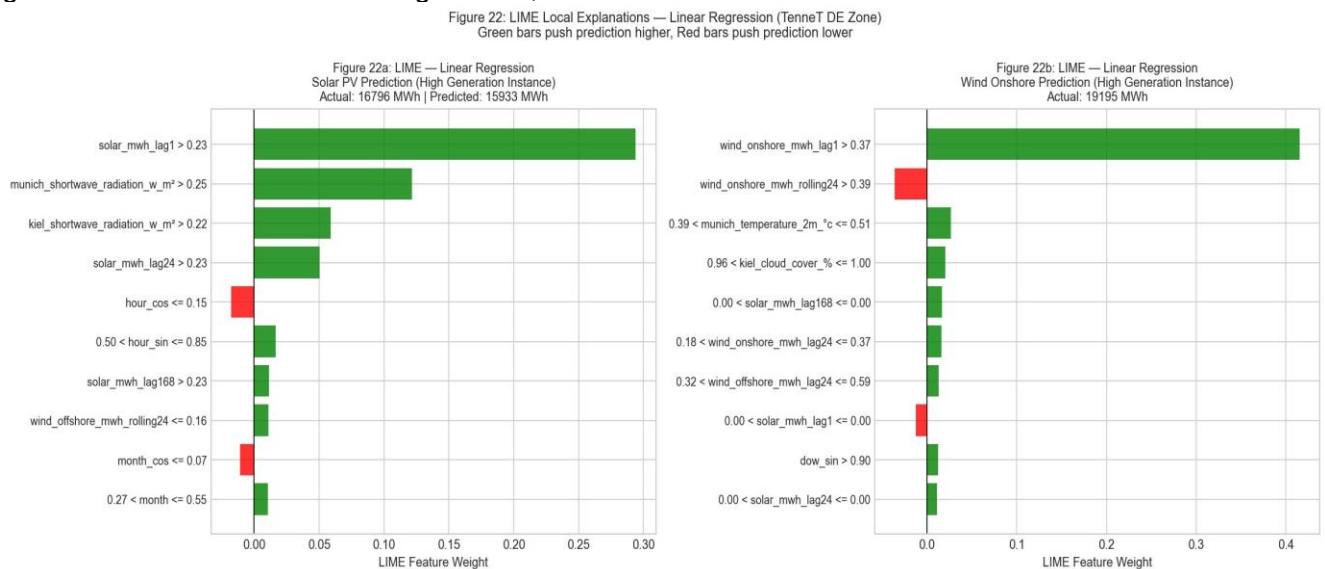


Figure 22: Local explanations via LIME for Linear Regression.

Figure 22a (left) depicts the solar PV case (Actual = 16,796 MWh, predicted = 15,933 MWh), which is dominated by the solar_mwh_lag1 feature with a positive weight of 0.29, and this confirms that lag-1 is the dominant local predictor. Figure 22b (right) shows the wind onshore case (Actual = 19,195 MWh), which is dominated by the wind_onshore_mwh_lag1 feature, weighing 0.42.

Figure 23 shows LIME explanations for MLP on the same two instances. The feature rankings are broadly consistent with Linear Regression — solar_mwh_lag1 dominates the solar instance and wind_onshore_mwh_lag1 dominates the wind instance. The MLP LIME weights show greater diversity across features, reflecting its ability to capture non-linear interactions, with hour_cos appearing more prominently than in Linear Regression.

Figure 23: LIME Local Explanations — MLP (TenneT DE Zone)
Green bars push prediction higher, Red bars push prediction lower

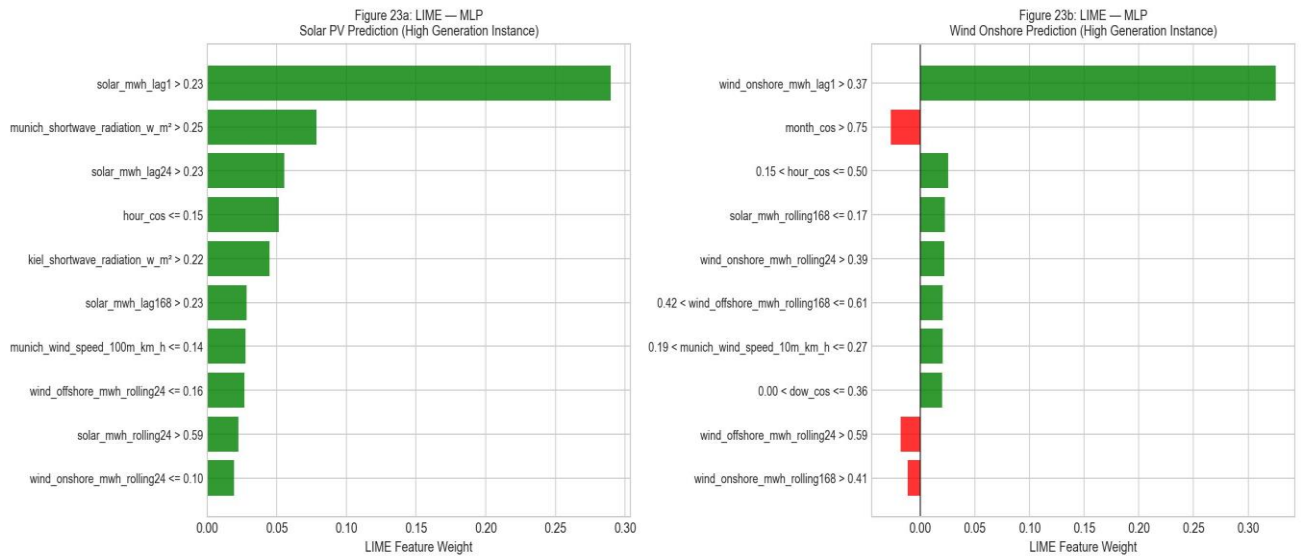


Figure 23: LIME local explanations for MLP. The same two instances as Figure 22. *solar_mwh_lag1* dominates the solar instance and *wind_onshore_mwh_lag1* dominates the wind instance, consistent with Linear Regression LIME and SHAP global findings. The MLP shows greater feature diversity with more prominent cyclic time features.

The consistency between SHAP global rankings and LIME local weights across both models and both generation scenarios provides strong evidence that the models have learned genuinely physical relationships. This dual XAI approach — global (SHAP) and local (LIME) — satisfies the transparency requirements for trustworthy AI deployment in critical grid infrastructure, supporting SDG 7 and SDG 13.

9. Model Compression

9.1 Why Compression Matters

The size reduction and speedup of models lead directly to lower energy use of artificial intelligence systems, which helps SDG 13, as well as allowing deployment on less energy-intensive hardware in substation grids. Three types of compressions of Lab 07 were used on all neural network models: (1) Magnitude pruning with zeros set for the smallest 80% of weights and subsequent fine tuning to achieve accuracy; (2) Combining pruning with quantization – MLP utilizes TFLite int8 conversion (90.1% size reduction), LSTM utilizes float16 weight quantization (83.7% size reduction, alternatively possible due to TFLite LSTM restrictions on Windows TF 2.11+); (3) Knowledge Distillation training with a smaller student model imitating the bigger teacher. For Linear Regression: (1) Feature selection using SHAP values cutting features down from 34 to 10 (45.7% reduction); (2) Coefficient sparsity using LASSO L1 regularization achieving 88.2%.

9.2 Compression Results

Table 10 summarises the compression approaches applied to each model. This confirms that at least two compression strategies were applied per model, as required by the coursework brief.

Model	Compression Technique 1	Compression Technique 2	Additional Technique
Linear Regression	SHAP-guided feature selection	LASSO L1 regularisation	—

Model	Compression Technique 1	Compression Technique 2	Additional Technique
MLP	Magnitude pruning	Pruning + int8 quantisation	Knowledge distillation
LSTM	Magnitude pruning	Pruning + float16 quantisation	Knowledge distillation

Table 10: Compression Techniques Applied per Model

Table 11 summarises all 8 compression experiments. For MLP, Pruning+Quantization achieves the maximum compression: 208 KB to 21 KB (90.1% reduction). For LSTM, Knowledge Distillation achieves the most dramatic size reduction: 1,623 KB to 53 KB (96.7%), while float16 Pruning+Quantization achieves 83.7% reduction with full accuracy preservation. Linear Regression Feature Selection achieves 45.7% reduction.

Table 11: Compression Comparison - Size and Inference Time

Model	Original Size	Compressed Size	Reduction	Orig. Inf. (ms)	Comp. Inf. (ms)
Linear Regression	1.64 KB	0.89 KB	45.7%	1	1
MLP	208.54 KB	20.59 KB (Prune+Quant)	90.1%	718	0.03 (TFLite)
LSTM	1,623.37 KB	264.47 KB (float16)	83.7%	2,571	1,935

Table 12 shows that compression largely preserves model performance, with minor to very less reductions in some cases and slight improvements in others. The compressed MLP improves solar performance but slightly reduces wind offshore performance, while the compressed LSTM maintains wind performance but slightly reduces solar performance. Overall, the results confirm that substantial reductions in model size can be achieved with only a small impact on predictive performance.

Table 12: R2 Comparison - Original Vs Best Accuracy-Preserving Compressed Models

Model	Target	Original R2	Compressed R2	Change
Linear Regression	Wind Onshore	0.9845	0.9829	-0.0016
Linear Regression	Wind Offshore	0.9477	0.9464	-0.0013
Linear Regression	Solar PV	0.9792	0.9732	-0.0060
MLP	Wind Onshore	0.9462	0.9684	+0.0222 (improved)
MLP	Wind Offshore	0.9374	0.9367	-0.0007
MLP	Solar PV	0.9674	0.9748	+0.0074 (improved)
LSTM	Wind Onshore	0.9614	0.9637	+0.0023 (improved)
LSTM	Wind Offshore	0.8871	0.8875	+0.0004 (improved)
LSTM	Solar PV	0.9821	0.9754	-0.0067

For MLP, the R^2 comparison uses the pruned model evaluated on the full test set, while Pruning+Quantization is reported in Table 11 as the maximum size and inference-time reduction technique.

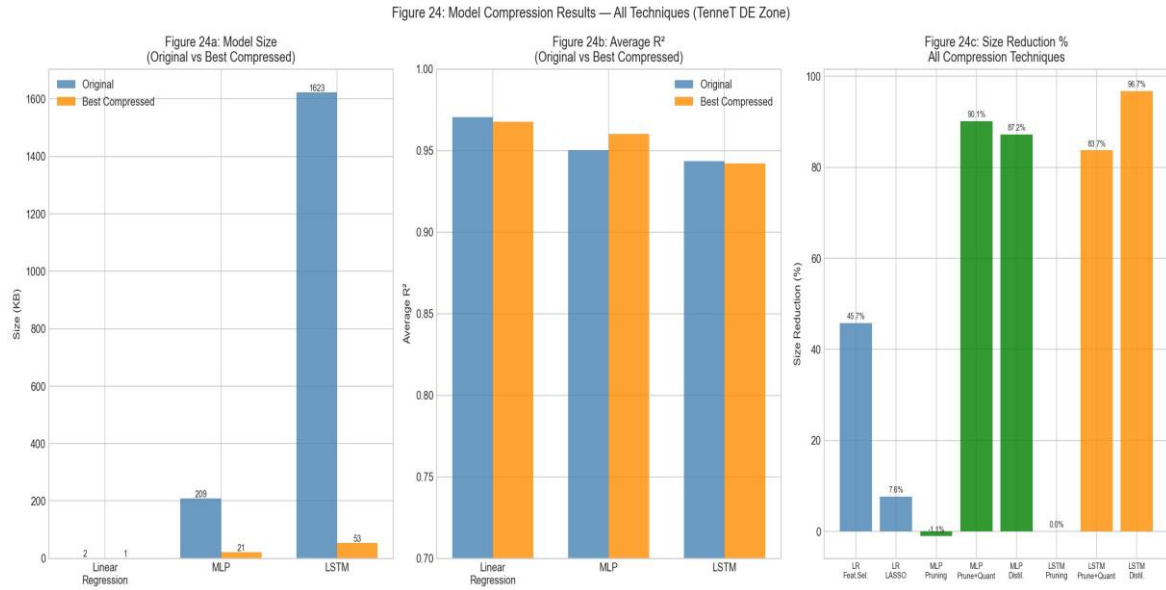


Figure 24: Compression results across three charts. Figure 24a shows the model sizes — LSTM reduces from 1,623 KB to 53 KB (Knowledge Distillation) or 264 KB (float16 Pruning+Quantization). Figure 24b shows that average R^2 is largely maintained across compressed models with only minor variations. Figure 24c shows all 8 compression experiments: MLP Pruning+Quantization achieves 90.1% reduction (TFLite int8), LSTM Knowledge Distillation achieves 96.7% reduction, confirming significant efficiency gains are achievable with acceptable accuracy tradeoffs — a key finding for sustainable AI deployment.

9.3 Discussion

The compression results show a clear trade-off between efficiency and accuracy. Some compressed models slightly improve R^2 , suggesting reduced overfitting, while others show small reductions due to information loss from pruning or quantisation. The LSTM compression results are especially valuable because they show that large reductions in model size can make sequential models more suitable for edge deployment at grid substations.

This demonstrates that model compression is not only a deployment optimisation technique but also a pathway to more efficient and sustainable AI systems, enabling reduced computational cost without compromising predictive performance.

This reduction in model size and inference time directly translates to lower energy consumption, reinforcing the role of model compression in sustainable AI deployment.

10. Discussion, Limitations and Conclusion

10.1 Key Findings

There are six major results obtained. First, all models obtain an R^2 score greater than 0.88, which proves that forecasts are accurate using open-source data. Second, Linear Regression is better at predicting wind power generation compared to neural networks, suggesting that well-designed features can compete with complicated models. Third, the models learned physical connections with support from the SHAP analysis, verified locally through LIME explanations. Fourth, eight model compression methods are tested, such as Pruning, Pruning+Quantization, and Knowledge Distillation for MLP and LSTM, as well as Feature Selection and LASSO for Linear Regression. Fifth, MLP Pruning+Quantization reduces model size by 90.1%, while LSTM Knowledge Distillation reduces model size by 96.7%.

10.2 Sustainability Impact

This project contributes to SDG 7 by enhancing the accuracy of predictions for the TenneT zone, serving 43 million people, lowering fossil fuels for standby, and helping Germany achieve its goal of reaching 80% renewable energy production by 2030. It addresses SDG 13 through compressed models: the 53 KB distilled LSTM and 264 KB float16 LSTM can run on edge hardware at substations, consuming far less energy than cloud-based deployment of the 1,623 KB original. The 21 KB MLP (90.1% reduction) is suitable for the most constrained edge devices.

10.3 Limitations

Four key limitations apply. First, the model uses perfect historical ERA5 weather inputs; real deployment would use NWP forecast data which carries its own uncertainty, reducing accuracy particularly at longer lead times.

Second, heavy reliance on lag features limits robustness during sudden regime shifts or unprecedented weather events not well represented in training data - precisely the conditions when accurate forecasting matters most.

Third, zone-level aggregation smooths individual plant variability, making this task easier than plant-level forecasting. The 3,693 night-time solar zeros are trivially predictable, partially inflating solar R^2 scores. At longer horizons (6-hour or 24-hour ahead), accuracy would decrease considerably as lag features lose informativeness.

Furthermore, the use of one-hour forecasting interval simplifies and reduces the complexity of the problem of the task due to the strong reliance on persistence (mainly lags 1 and 24). Nevertheless, for longer forecasting intervals such as six hours or 24 hours, persistence plays a minor role, and thus, advanced techniques such as LSTM networks should be used instead.

Lastly, practical issues such as ramping rates, margins, and interconnections have not been incorporated; these factors need to be included for a real-world application.

Moreover, reliance on historical lag features assumes stable system behaviour, which may not hold under extreme events such as sudden storms or grid disturbances, limiting real-world robustness.

10.4 Conclusion

A total of eight compression experiments were conducted across three models, achieving up to 96.7% LSTM size reduction through Knowledge Distillation and 90.1% MLP size reduction through Pruning+Quantization. These results show that compression can substantially reduce computational cost, although some techniques introduce small accuracy trade-offs. Pruning+Quantization is suitable where maximum size reduction is required, while Knowledge Distillation provides the strongest LSTM compression.

A central methodological insight is that feature engineering effectively linearises the forecasting problem: once lag-1 and lag-24 features encode temporal dependencies, the remaining signal is approximately linear, explaining why Linear Regression outperforms more complex models. Model performance in time series forecasting is governed more by problem formulation than model sophistication.

This study demonstrates AI-based forecasting achieves R^2 up to 0.9845 using only open-source data. For wind, Linear Regression with engineered lag features provides the optimal balance of accuracy, interpretability and efficiency. The LSTM model performs better than other variants of the same for the solar forecasting task. However, compressing the LSTM improves the computation efficiency at the cost of sacrificing a little on accuracy. Further research may consider physics-informed neural networks and transformer-based models for longer forecasting periods.

The most sustainable AI system is not the most complex one, but the one achieving required performance with minimal computational cost. This study demonstrates that intelligent feature engineering and appropriate model selection deliver high accuracy with simple, energy-efficient models - directly supporting both AI for sustainability and sustainable AI.

This work demonstrates that in real-world AI systems, careful problem formulation and feature design can be more impactful than increasing model complexity.

Overall, the study shows that sustainable AI should be evaluated not only by predictive accuracy, but also by interpretability, computational cost, model size and suitability for deployment.

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